

MULTIPLE CHOICES QUESTIONS

Q4	Q5	Q6	Q7
Q13	Q14	Q15	Q16
Q17	Q18		

Part 2: Essay Questions (20 marks)

QUESTION #2

Write instructions to test bit-5 in register AL, if it is one reset bit-1 otherwise set bit-2

Write a procedure to store 'x' into 1 KB memory locations defined as below:
DB 1024 DUP(?)

Using BIOS interrupt, write assembler codes to clear screen and set cursor to position (10, 20).

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1. To transfer value (6F17) 16-BIT data from microprocessor to output port with address 3F0H the following instructions can be used:

- ☒ A. MOV DX, 03F0H; MOV BX, 6F17H; OUT DX, BX
 B. MOV DX, 03F0H; MOV AX, 6F17H; OUT DX, AX
 C. MOV DX, 03F0H; MOV AX, 6F17H; OUT DX, AL
 D. MOV DX, 03FCH; OUT DX, AL

2. The interrupt generated by the execution of an interrupt instruction INT is

- A. NMI interrupt B. software interrupts C. external interrupt D. Interrupt-in-interrupt

3. The physical address of interrupt vector associated with INT 16 is:

- A. 00042H B. 00040H ☒ C. 00064H D. 0003CH

4. The size of interrupt vectors table is ----

- ☒ A. 4 bytes B. 1024 bytes C. 64 bytes D. 128 bytes

5. The binary value of AL and carry flag (CF) after the following instructions have executed :

MOV AL, 0ACH

SAR AL, 1

- A. AL=01010101B; CF=1 ☒ B. AL=11010110B; CF=0
 C. AL=01010100B; CF=0 D. AL=11010110B; CF=1

6. The binary value of AL and CF after the following instructions have executed is ----

MOV CL, 4

MOV AL, 7FH

ROL AL, CL

- A. AL=11110111B; CF=0 B. AL=11110111B; CF=1
 C. AL=01111111B; CF=1 D. AL=01111111B; CF=0

7. In the following instruction sequence, show the value of AL, in binary:

MOV AL, 3FH

NEG AL

XOR AL, FFH

- ☒ A. AL= 00111110B B. AL= 01011111B C. AL= 01011110B D. AL= 01011111B

8. If there is a carry from lowest 4 bit during addition, _____ flag sets.

- ☒ A. Auxiliary carry B. Carry C. Over flow D. Sign

9. The result of multiplication instruction MUL BX is stored in register pair

- ☒ A. DX:AX B. AX:DX C. CX:AX D. AX:CX

10. Find the content of register AL after execution of the following code:

MOV AL, 11

SHR AL, 1

ADC AL, 1

- A. 7 ☒ B. 5 C. 4 D. 3

11. Find the content of memory location X after execution of the following code

X DB ?

MOV AX, 11

MOV BL, 3

DIV BL

OR AH, 30H

MOV X, AH

- ☒ A. X='2' B. X=0 C. X='0' D. X=2

12. Find the content of AH registers after execution of the following code:

MOV AX, 14

MOV BL, 4

DIV BL

XOR AH, 0FH

- A. D0H B. C0H C. 0DH D. 0CH

13. LOPPE instruction repeat a group of instruction while

- A. (CX=0) & (ZF=0) B. (CX=0) & (ZF≠0) C. (CX≠0) & (ZF=1) D. (CX≠0) & (ZF=0)

14. The value of CX and AL after executing the following code is:

```
MOV AL, FFH
MOV CX, 6
L1: ADD AL, 1
    LOOPZ L1
```

- A. CX=4; AL=2 B. CX=2; AL=3 C. CX=5; AL=1 C. D. CX=3; AL=6

15. To compare two strings in equality we use prefix _____ before instruction CMPSB:

- A. REP B. REPNE C. REPE D. REPZ

16. The equivalent instructions of STOSB; assuming DF flag =0:

- A. MOV AL,[SI]; INC SI B. MOV AL,[SI]; DEC SI
C. MOV [DI],AL; INC DI D. MOV [DI],AL; DEC DI

17. What will be appears on the screen after executing the following code:

```
STR DB "CDEFGHK", 0DH, "AB"
    DB 'XYZ', 'S'
MOV AH, 9
MOV DX, OFFSET STR
INT 21H
```

- A) CDEFGHAB B) ABXYZH C) ABCDEFGH D) ABXYZHK

18. The values of array Y after executing the following code is:

```
X DB 'ABC+CDK'
Y DB 5 DUP('*')
CLD
MOV SI, OFFSET X
MOV DI, OFFSET Y
MOV CX, 3
REP MOVSB
MOV AL, '#'
STOSB
```

- A) Y=ABC+CDK B) Y=ABC*# C) Y=ABCH*CDK D) Y=ABCH*

19. The value of Y after executing the following code is:

```
X DB 4,7,8
Y DB ?
MOV BX,OFFSET X
MOV CX, 2
MOV AL,0
L: CALL XXX
    LOOP L
    MOV Y, AL
    HLT
XXX PROC
    MOV AH, [BX]
    INC BX
    ADD AL, AH
    RET
```

- XXX ENDP A) 11 B) 04h C) 0 D) 9

20. In asynchronous serial data transmission, assume The data format is: 8 data bit, 1 start bit, 1 stop bit and even parity, What is the speed of transmission (baud rate) if the device transmit 20 characters per second?

- A) 220 B) 20 C) 180 D) 200

- Write an assembler code to reverse string X in order; use LODSB and STOSB strings; where
X DB 'ABCDEK++*'

QUESTION#3

Interface 8-bit input port (74LS245) to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X defined as byte. Assume the address of the port is 0BFFH.

- Draw block circuit diagram circuit, showing address decoding logic, and control signals
- Write a program to read the status of switches and store the result into X memory location

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